

Embedded AVR Course

✓ → C Language → 30 hrs



✓ → Basics of Computer Architecture 6 hrs

✓ → // of Digital input output DIO 6 hrs

✓ → Interrupt 3 hrs

✓ → Timers → (Normal, pwm, ICA, wDT, CTC) ③ 9 hrs

✓ → ADC → 3 hr

✓ → Communication protocols (UART, I2C, SPI)

مجموع

66 hr

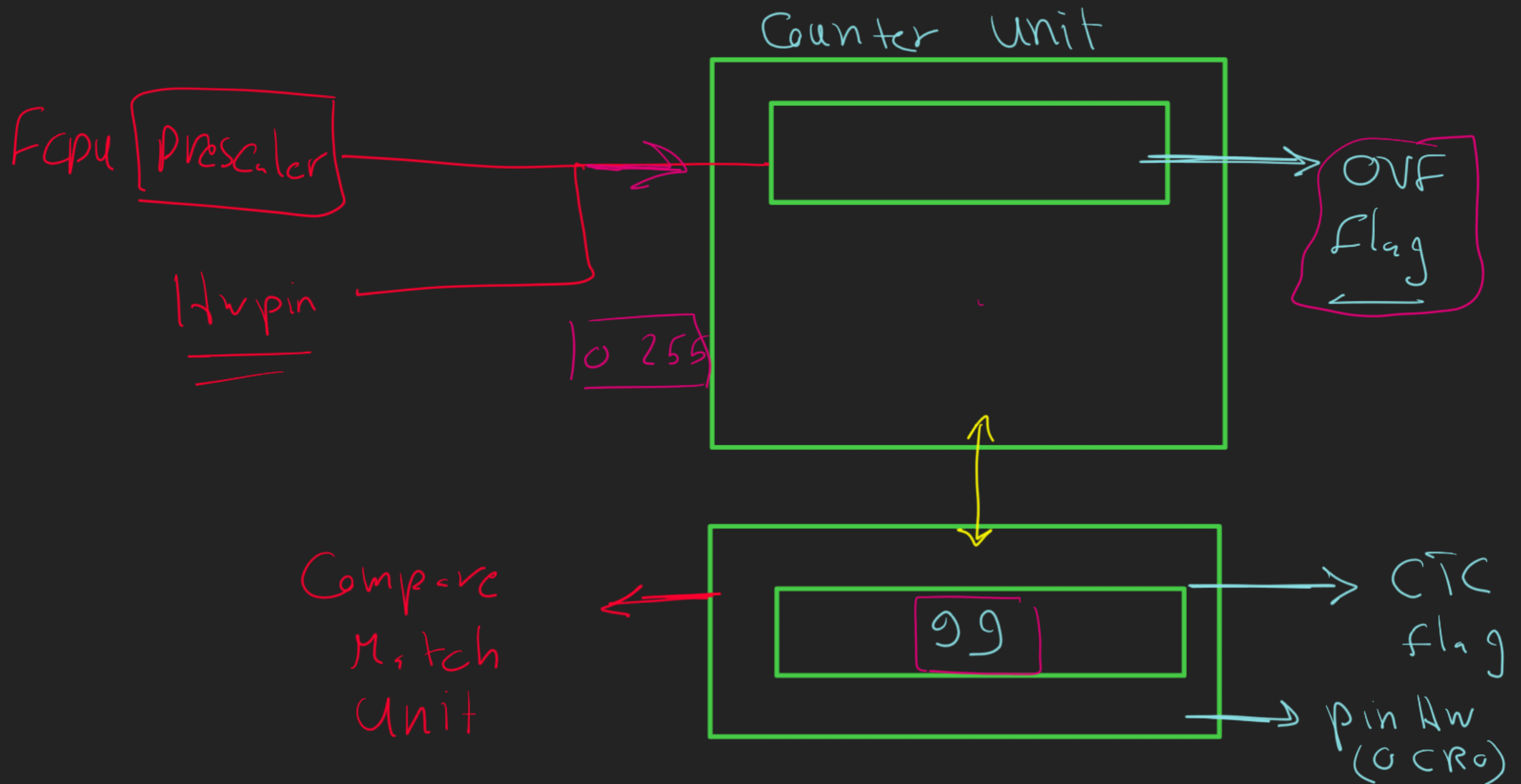
10

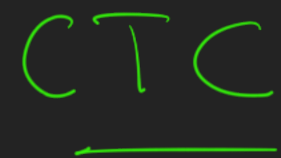
9 hrs

36

CTC Mode

→ Clear on Compare Match




$$\text{ISR}(\text{CTC})$$

A hand-drawn diagram illustrating a set. On the left, a large curly brace $\{$ is drawn. To its right, there are five horizontal lines representing elements of the set. The first line is a simple horizontal line. The second line is a horizontal line with a small circle at its right end. The third line is a horizontal line with a small circle at its right end. The fourth line is a horizontal line with a small circle at its right end. The fifth line is a horizontal line with a small circle at its right end. On the far right, a large curly brace $\}$ is drawn, closing the set.

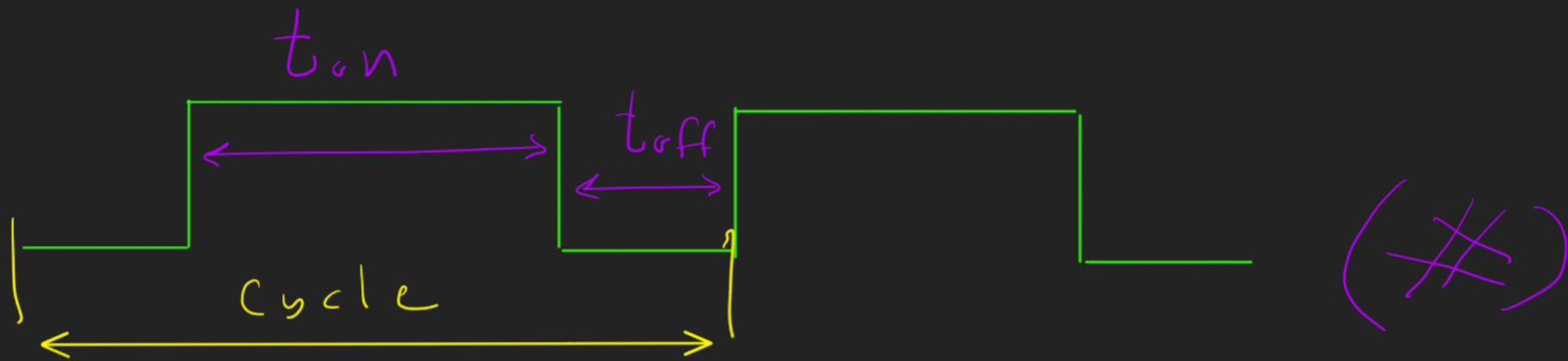
* PWM $\rightarrow$ pulse width Modulation

[1] Freq $\rightarrow$ Const

[2] Time on, Time off

$$\text{Time Total} = T_{\text{on}} + T_{\text{off}}$$

[3] Amplitude $\rightarrow$ Could be change.



AC DC ? $\rightarrow$ Max
RMS = (Root Mean Square)

$5 + 10 = 15 \text{ sec}$

$\frac{5}{15} = \frac{1}{3} \text{ } 33\%$

$$\underline{\text{Duty Cycle}} = \frac{T_{on}}{\text{Time total}} = T_{on} + T_{off}$$



$$R_{ms} = \underline{A} \sqrt{\underline{\text{duty cycle}}} = \underline{5} \sqrt{\underline{0,333}} =$$

Digital 0 $\rightarrow$ 1
 $\rightarrow$ output

2.88 V

PWM

→ output

Digital pin

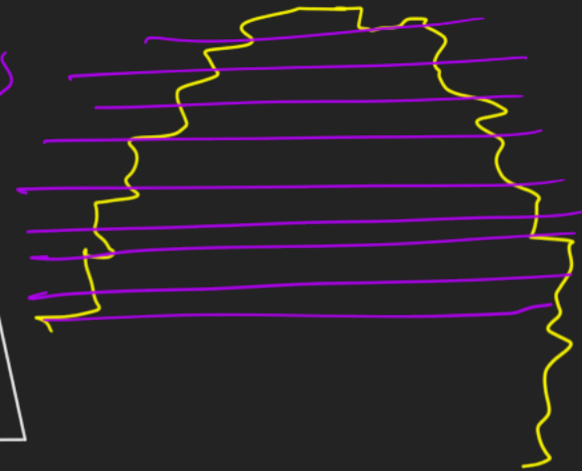
Analog ADC

⇒ Input →

Digital pin

PWM = DAC

Levels



RMS

How to PWM?

while (1)

{

setpin (High);

delay (300);

→ setpin (Low);

delay (700);

~~time ()~~

time ()

Not efficient

Duty cycle

$$= \frac{300}{1000}$$

$$= 0,3 =$$

30%

Timer CTC, Normal Mode

ISR (CTC) $\Rightarrow$ 10 Msec

{ static unsigned char counter = 0;

counter ++;

if (counter == 30)

{ set pin low

} else if (counter == 100)

{ set pin high

30%

~~30%~~

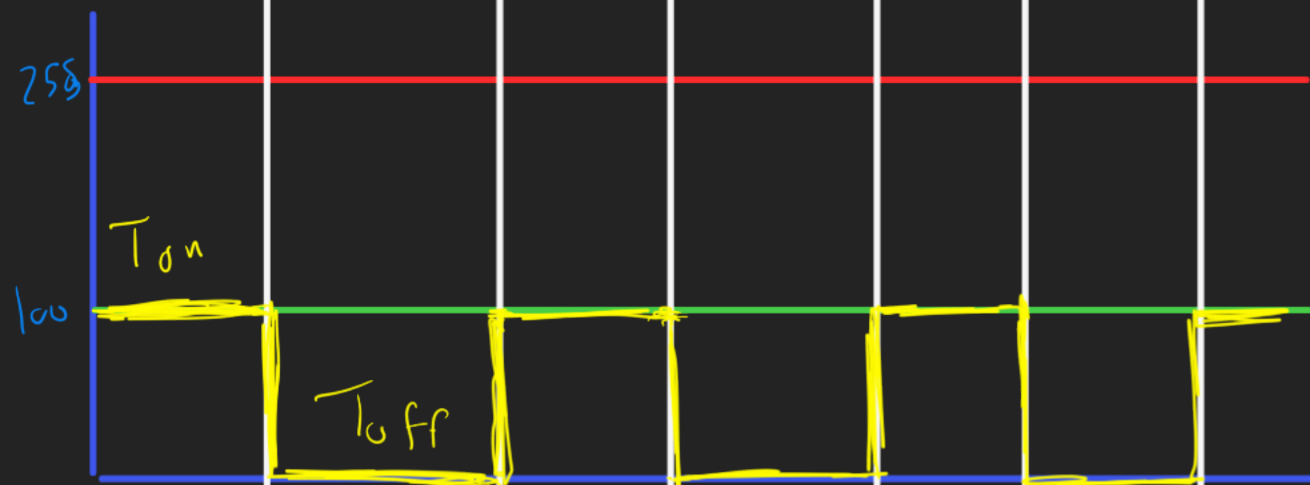
70%



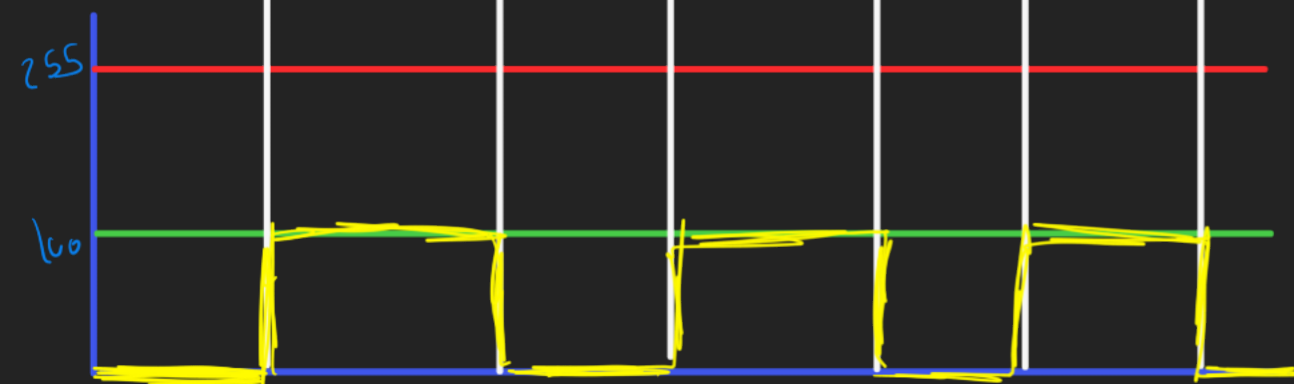
(1) Clear
CoCo pin
on Compar

Duty

$$= \frac{100}{255} = 40\% \text{ Match}$$
 Non inverting
 $T_{off} > T_{on}$



(2) Set oCo
pin on Compar
Match
inverting



(2)

100 Compar
 Duty cycle = $\frac{255 - 100}{255} = 60\%$